

Algoritmo $\mathcal{I}$ — Paso 3: generación de las restricciones

1 $\mathcal{I}(\Gamma \mid \text{True}) = (\text{Bool} \mid \emptyset)$

2 $\mathcal{I}(\Gamma \mid \text{False}) = (\text{Bool} \mid \emptyset)$

3 $\mathcal{I}(\Gamma \mid x) = (\tau \mid \emptyset) \quad \text{si } (x : \tau) \in \Gamma$

4 $\mathcal{I}(\Gamma \mid \text{if } M_1 \text{ then } M_2 \text{ else } M_3) = (\tau_2 \mid \{\tau_1 \stackrel{?}{=} \text{Bool}, \tau_2 \stackrel{?}{=} \tau_3\} \cup E_1 \cup E_2 \cup E_3)$
donde $\mathcal{I}(\Gamma \mid M_1) = (\tau_1 \mid E_1)$
 $\mathcal{I}(\Gamma \mid M_2) = (\tau_2 \mid E_2)$
 $\mathcal{I}(\Gamma \mid M_3) = (\tau_3 \mid E_3)$

5 $\mathcal{I}(\Gamma \mid M_1 M_2) = (X_k \mid \{\tau_1 \stackrel{?}{=} (\tau_2 \rightarrow X_k)\} \cup E_1 \cup E_2)$
donde X_k es una incógnita fresca
 $\mathcal{I}(\Gamma \mid M_1) = (\tau_1 \mid E_1)$
 $\mathcal{I}(\Gamma \mid M_2) = (\tau_2 \mid E_2)$

6 $\mathcal{I}(\Gamma \mid \lambda x : \tau. M) = (\tau \rightarrow \sigma \mid E)$
donde $\mathcal{I}(\Gamma, x : \tau \mid M) = (\sigma \mid E)$

7 $\mathcal{I}(\Gamma \mid \text{zero}) = (\text{Nat} \mid \emptyset)$

8 $\mathcal{I}(\Gamma \mid \text{succ}(M)) = (\text{Nat} \mid \{\tau \stackrel{?}{=} \text{Nat}\} \cup E) \quad \text{donde } \mathcal{I}(\Gamma \mid M) = (\tau \mid E)$

9 $\mathcal{I}(\Gamma \mid \text{pred}(M)) = (\text{Nat} \mid \{\tau \stackrel{?}{=} \text{Nat}\} \cup E) \quad \text{donde } \mathcal{I}(\Gamma \mid M) = (\tau \mid E)$

10 $\mathcal{I}(\Gamma \mid \text{isZero}(M)) = (\text{Bool} \mid \{\tau \stackrel{?}{=} \text{Nat}\} \cup E) \quad \text{donde } \mathcal{I}(\Gamma \mid M) = (\tau \mid E)$